

W03 BASIC WATER MANAGEMENT | P

Intent : Implement protocols to reduce risk of water quality loss and *Legionella* colonization

Summary : This WELL feature requires projects to proactively test drinking water and to manage recirculating hot water systems against *Legionella* colonization.

Issue : All water systems require some degree of validation to ensure that the health targets are met under its operational conditions, for which sampling is an effective verification tool ¹. *Legionella* bacteria is naturally present in waters at low concentrations, but it may colonize recirculated water systems and can cause lung disease and even death if contaminated water aerosols are inhaled.² Legionnaire's disease affects especially the immunocompromised, smokers and those over age 50. ² Assets commonly vulnerable to *Legionella* infestation include domestic potable and hot water systems, cooling towers, humidifiers, misters, decorative fountains, spas and hot tubs.²

Solutions : Basic management for water quality ensures that the water sourcing, treatment and delivery operates as designed. Turbidity, residual chlorine and pH monitoring help control basic chemical and microbiological water characteristics and, if trends are detected, may inform that actions may be needed to protect the quality of the water, such as changing a filter, check the building's pipes for leaks or inquire with the city for major works in the area. Management for minimizing *Legionella* requires a thorough and quantitative risk assessment of buildings' water assets, identifying locations where control is required, and a well-documented maintenance and operations program.³ Implementing a proper *Legionella* management plan should reduce the risk of exposure to pathogenic bacteria.

PART 1 MONITOR CHEMICAL AND BIOLOGICAL WATER QUALITY

For All Spaces except Dwelling Units:

1: Drinking water quality report

The following requirements are met:

- a. The following water parameters are sampled at drinking water dispensers in occupiable spaces at intervals of no less than once per year:
 1. Turbidity.
 2. pH.
 3. Residual (free) chlorine.
 4. Total coliforms, only if residual chlorine is below detection limits
- b. Tests are required at 5% of drinking water dispensers, up to a maximum of four tests.
- c. The water quality results are submitted annually through the WELL digital platform.

Note :

On-going testing does not need to be performed by a WELL Performance Testing Agent.

PART 2 IMPLEMENT LEGIONELLA MANAGEMENT PLAN

For All Spaces:

1: Legionella plan development

The project provides a Legionella management plan that meets the following requirements:

- a. Addresses hot water systems, cooling towers, decorative fountains and any other devices or spaces under control of the project where water is recirculated and aerosolized.
- b. Includes the items listed below:³
 1. Determination of roles for Legionella management in the building, distinguishing those under project control from those that may be the responsibility of building management or other parties.
 2. Water system inventory and process flow diagrams of systems within the project boundary.
 3. Hazard analysis of water assets within the project boundary. If the project does not operate the building hot water supply system (e.g., boilers, heaters, pumps or hot water risers), then an explanation of the building-wide Legionella management policies (if any) and how they influence risk is included.
 4. A list of monitoring actions for relevant variables (e.g., temperature or residual chlorine), performance limits associated with these variables, and corrective actions when variables exceed such limits.
 5. A list of critical control points (locations where actions to maintain relevant variables listed in (4) within performance limits are applied) within the project boundary.
 6. Verification and validation procedures for evaluating the suitability and proper implementation of the management plan. A Legionella sampling schedule is included if projects have operational control over cooling towers and spas.
 7. Protocols for documenting results of monitoring activities and corrective actions. If sampling for Legionella is planned, results are included.

2: Legionella plan implementation

The following requirement is met:

- a. Project submits annually through the WELL digital platform documentation of monitoring results, corrective actions and *Legionella* sample results (if any) as stated in the Legionella management plan.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. US Centers for Disease Control & Prevention. Legionella (Legionnaires' Disease and Pontiac Fever). <https://www.cdc.gov/legionella/index.html>. Published 2019. Updated April 30, 2018. Accessed December 20, 2019.
3. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 188-2018: Legionellosis: Risk Management for Building Water Systems. In.